# Tag & Track

Conner Museum of Vertebrates

For: Jessica Tir and Katherine Corn
Developers: Martin Hundrup, Shawn Will, and Sarah Carley

# Introduction to Tag & Track

- **Objective:** Our objective is to make an iPad app that allows the employees at the Conner Museum to more easily perform their loan tracking.

- **Purpose:** The current way the Museum does things is all on paper, which can get hard to keep track of as well as tedious to work with. Our app aims to make this part of their job less tedious.

- **Scope:** The scope for this project includes loading specimen from arctos, allowing the creating and saving of loans, a way to track what shelf specimen are on, a way to export loan data, as well as a centralized QR code system to make everything quickly scannable.

# Team Introduction

- Introductions - Tag and Track Team
  - Martin Hundrup - Team Lead and CS Major, Back End
  - Sarah Carley - CS Major, Front End
  - Shawn Will - CS Major, Database
- Objectives
  - Finish all remaining promised features
  - Fully implement tests
  - Handle all received feedback
  - Finalize our documentation

# Sprint Objectives

## Defined Final Goals

- Complete all user stories
- Deploy application to client
- Address client feedback

## Connection to Vision

- These goals align with the team's desire to deploy a well-working application to the client

## End-Of-Project Milestones

- The application will be ready for real use in loan management for the Conner Museum

# Feature Implementation (Martin)

## Feature Overview

- Data Export/Import
  - Allows for the exporting of the entire DB to a csv file, which can be reimported.
- Scanning Containers
  - Scanning a container will add all checked-in items in the container to an ongoing loan creation
- Manual Item Entry
  - A new page that allows any checked-in item in the DB to be added to an ongoing loan creation

## Testing and Validation

- Unit and Functional Testing

# Feature Implementation (Martin)

## Feature Overview

- Individual Specimen Check In
  - Allows for individual items in a loan to be checked in. The check-in date is stored, and the item can then be checked out in future loans.
- Settings
  - Added DB import/export and a DB reset functionality to the settings page.
- Specimen Loan History
  - Individual specimen loan history can be viewed on a specimen info page.

## Testing and Validation

- Unit and Functional Testing

# Feature Implementation (Shawn)

## Feature Overview

- For this Sprint, I was tasked with updating our Documentation, preparing our Presentation(s), and creating our LinkedIn video
  - Our Readme was updated to reflect the functionality of the app
  - Our final report was officially finalized and submitted
  - Our poster board was setup in preparation for the Senior Design Poster Event
  - Our LinkedIn video was filmed, edited, and posted

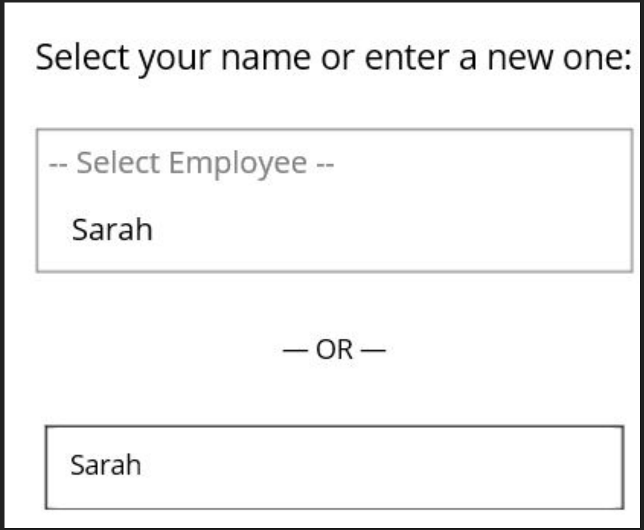
# Feature Implementation (Sarah)

## Feature Overview

- Made textboxes easier to see visually overall to address user feedback.
- Addressed potential memory leaks brought up by theme subscriptions.

## Testing and Validation

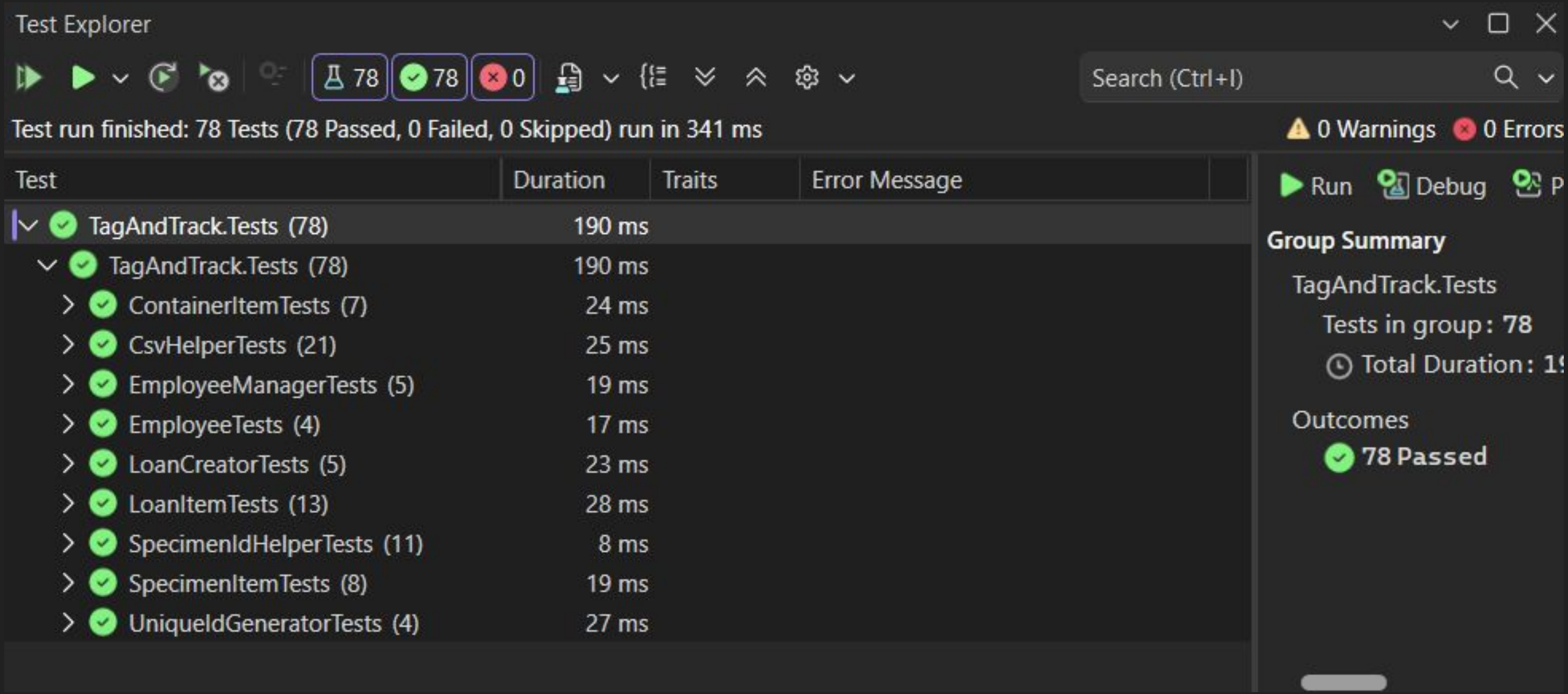
- Visual changes were tested manually by loading up the application.



# Live Demonstration

Demo

# Testing and QA



# Deployment of System

- Deployment Platform
    - The application will be deployed locally on the Museum's iPad with the use of an Apple Developer account.
- Process of Deployment
    - A team member went to the client on 4/21/26 and was successfully able to deploy the application with the client.
    - The application was manually loaded onto the museum's iPad via the development environment.
- System Access
    - Access to this system is native only to the museum's iPad. If the application were loaded elsewhere, the data would not be the same. (However, it can be exported and imported to another device)

# Deployment of System

- Client Feedback (Post Delivery)
  - The client was overall satisfied with the application
  - Feedback on UI elements and bug reports were given
- Post-Deployment Testing
  - User testing was performed over the weeks where the client used the application in their day-to-day
  - Unit tests were created on the backend that ensured correct functionality

# Kanban Overview and Contributions

[Kanban Board]

# MoM Evidence

MoM

# Documentation of Final Report

Walkthrough & Handoff of Design Docs

# Responsible Use of AI

## Usage

- AI was used at times during development of this project.

## Applications and Improvements

- Referenced at times to assist in coming up with approaches to code problems, improving efficiency of certain code by at least 20%.

## Limitations

- We would verify any code made by the AI and do thorough testing to ensure it was sound.

**We understand that final decisions and deliverables are our responsibility.**

# Sprint 6 Achievements and Challenges

## What went well

- Final deliverable has the requested features
- All user stories unrelated to Arctos were able to be completed before deployment
- Smaller fixes were quickly implemented when they were spotted

## Challenges Faced

- Lack of Apple developer account until recently limited testing time
- Certain aspects were planned later on due to the lack of an Arctos API key (csv import/export, manual database entry)

# Retrospective and Final Reflection

**Lessons Learned:**

- Developing for an Apple device requires additional resources
- If one way doesn't work out, there is always a fallback method that can be implemented

**Retrospective:**

**What went well**

- Completion of final goals
- Csv import/export
- Manual specimen entry to a loan
- Addressing feedback

**What can be Improved**

- If an Arctos API key could have been obtained, the data would be much more improved

**Client Feedback Actioned:**

- Some textboxes on the user interface were hard to see, so a border was used around them for visibility.
- There was a bug where the log-in feature wasn't working as intended.

# Conclusion & Handover

## Recap

- The project has been an overall success
- The application is well-developed and full of useful features for specimen/loan management
- The application is ready for deployment with the Conner Museum

## Handover Status

- This system has been handed over to the staff at the Conner Museum and will soon be used in their day-to-day activities.

# Thank you!